

Passive DS-TWR Known-Geometry Calibration

Static receive-only tag validation

UWB Module project

27 July 2026

Abstract

This report validates the project-specific Passive DS-TWR protocol after restoring a surveyed four-anchor geometry. Hardware-bias calibration reduces the live position-bias magnitude from 3.51 cm to 0.96 cm . The embedded solver improves from 4.12 cm to 3.33 cm 2-D RMSE. Re-solving the same calibrated observations with the surveyed geometry reaches 1.89 cm **RMSE** and 3.11 cm **P95** at $49.17\text{ complete frames/s}$. The final static position therefore improves. A direct before/after comparison also shows that the range calibration works very well, while the directed-observation calibration still interacts with the range correction and requires a joint refit.

1 Experimental setup

Four anchors formed a surveyed 3.000 m square and tag 1 was placed at its center:

Node	Coordinate [m]	Node	Coordinate [m]
A2	(0.000, 0.000)	A3	(0.000, 3.000)
A4	(3.000, 0.000)	A5	(3.000, 3.000)
Tag 1	(1.500, 1.500)	Placement tolerance	approximately $\pm 2\text{ mm}$

Both independent captures lasted 120 s and used Robust Rotating scheduling, 5 ms slots, a 1 ms frame gap, 1000 us RESP and FINAL delays, and UWB channel 5. The first capture used raw observations and raw piggybacked DS ranges. The second capture used the calibration derived from the first capture.

2 Position results

Processing path	Positions	Rate [Hz]	Bias X/Y [cm]	RMSE [cm]	P95 [cm]
Uncalibrated, autonomous geometry	3,362	28.00	+3.24/ -1.35	4.12	6.09
Calibrated, autonomous geometry	3,512	29.26	-0.66 / -0.70	3.33	5.66
Calibrated, surveyed geometry	5,901	49.17	-0.30 / -0.27	1.89	3.11

Calibration improves the live embedded RMSE by 19.2% and reduces the systematic position-bias magnitude by 72.7% . The HTTP snapshot collector keeps only the newest embedded position, so its $28\text{--}29\text{ samples/s}$ is a capture limit. Offline reconstruction of one solution per complete three-slot star shows the actual 49.2 frames/s position cadence.

Calibration effect	Before [cm]	After [cm]	Change
Live position RMSE	4.12	3.33	19.2% better
Position-bias magnitude	3.51	0.96	72.7% better
Mean absolute effective-range bias	5.63	0.17	96.9% better
Mean absolute directed-path bias	4.65	5.33	14.7% worse

The current calibration therefore helps the final position and nearly removes pair-range bias, but it does not yet independently calibrate every directed observation.

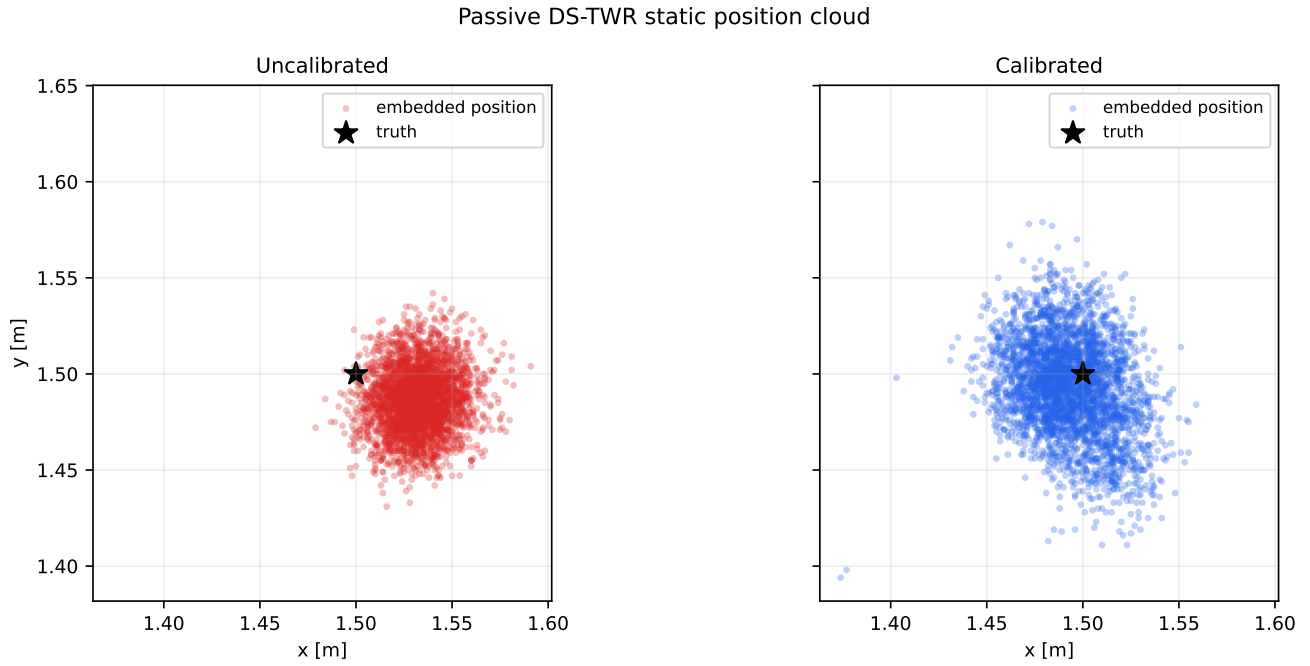


Figure 1: Embedded position clouds before and after calibration. The black star is the surveyed tag position.

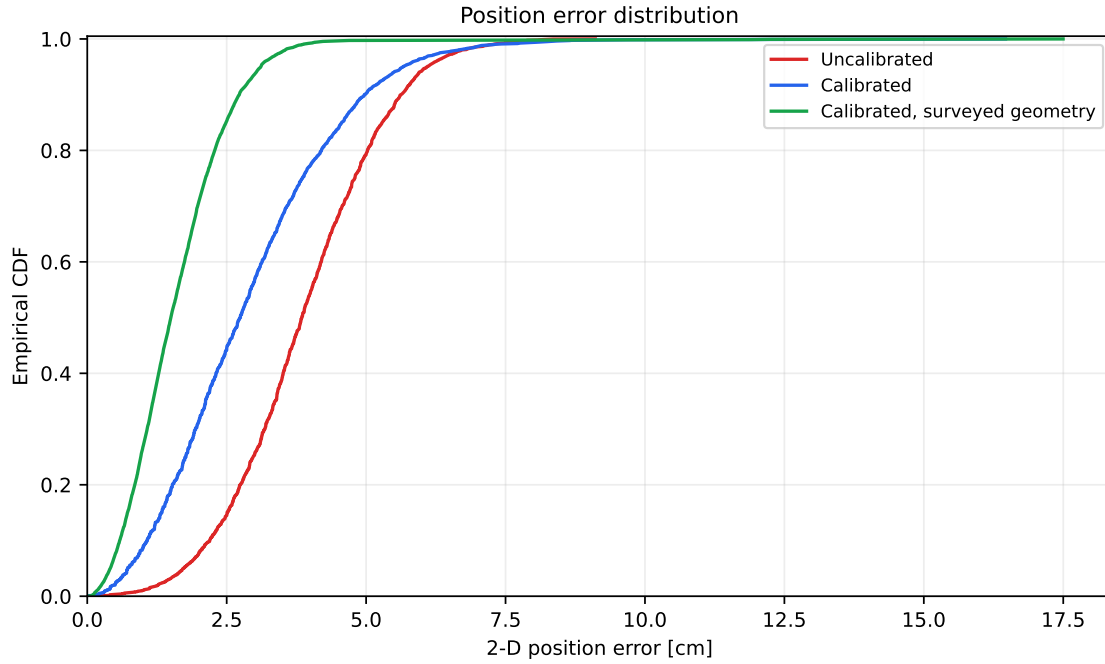


Figure 2: Empirical 2-D error distributions. Replacing only the autonomous geometry with surveyed coordinates exposes the quality of the calibrated passive observations.

3 Calibration model

The scalable directed-observation model is

$$e_{I \rightarrow R} = b_R - b_I,$$

where b_I and b_R are per-anchor biases. A2 fixes the arbitrary common offset. The first half of the uncalibrated observation capture produced:

Anchor	Observation bias	Anchor	Observation bias
A2	0 mm	A3	+34 mm
A4	-16 mm	A5	-55 mm

This uses only $N - 1$ independent values for N anchors and fits the antisymmetric component of all 12 directed paths. The piggybacked DS ranges use one additional correction per unordered anchor pair:

Pair	Bias	Pair	Bias	Pair	Bias
A2-A3	+71 mm	A2-A4	+56 mm	A2-A5	-49 mm
A3-A4	+67 mm	A3-A5	+66 mm	A4-A5	-29 mm

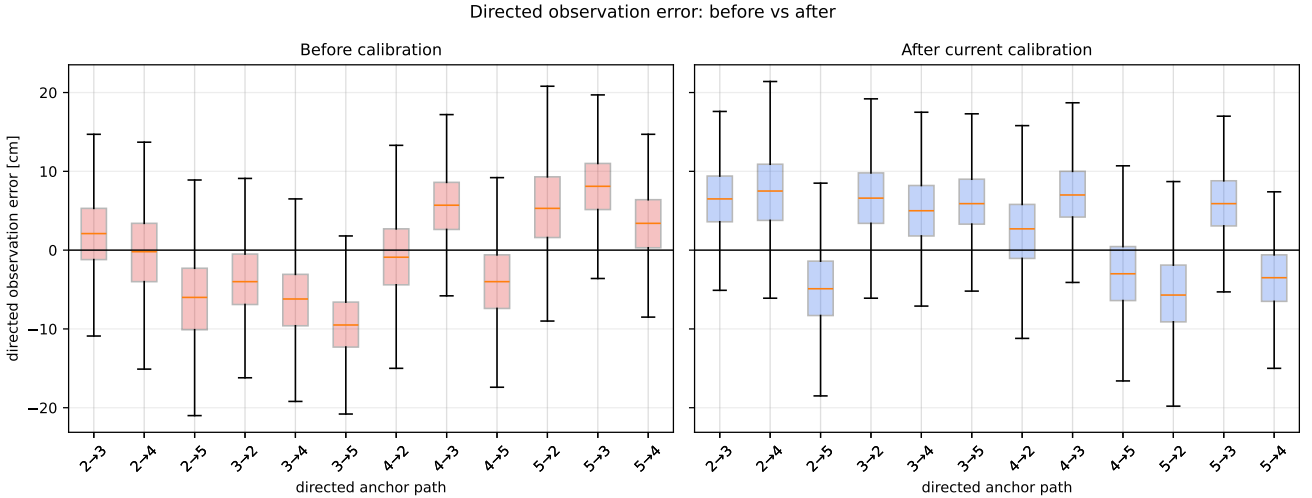


Figure 3: Direct before/after comparison on the same scale. The current combined calibration does not move every directed-path median toward zero.

4 Protocol health and interpretation

During the calibrated capture, anchor radio success ranged from 97.37% to 98.49%. The embedded calibrated distribution contains five estimates above 10 cm (0.142%), including two above 15 cm. They are short transients and remain included in every reported statistic and plot.

The uncorrected pair distances still fit a planar geometry with 1.99 cm range-residual RMS, but the resulting anchor coordinates are displaced by several centimeters. The direct comparison supports the following conclusions:

- CFO correction of the responder reply interval is working;
- piggybacked DS ranges clearly benefit from per-pair corrections;
- final static position RMSE and bias improve;
- the independent per-anchor observation fit is incomplete once the range correction is active;
- autonomous geometry and the observation/range interaction remain accuracy penalties.

The next implementation step is a joint observation fit performed after range correction, with an antisymmetric per-anchor term and, if needed, a symmetric per-pair observation term. Pair ranges should then be robustly filtered before they update geometry: a rolling median or Huber estimate, a physical-change gate, and a slower geometry-update cadence. Validation should cover several static tag locations and a dynamic trajectory. A single center position identifies the current bias, but cannot establish workspace-wide accuracy.

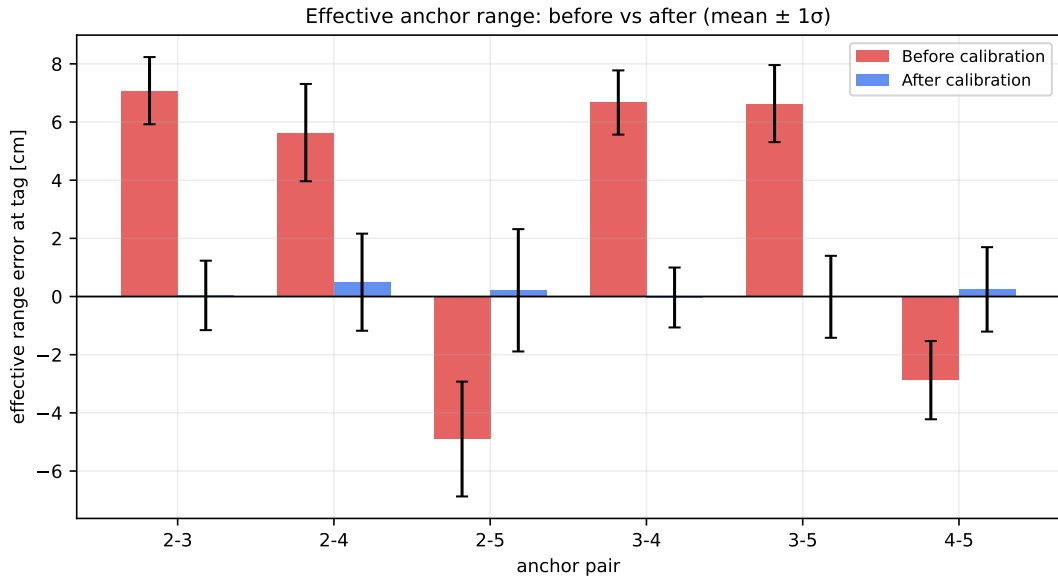


Figure 4: The effective anchor range used by the tag before and after calibration. Pair-range bias is almost completely removed.

5 Reproducibility

The repository preserves both compressed JSONL captures, metadata, wireless logs, SHA-256 manifests, CSV metrics, figures, and the deterministic analyzer:

```
python3 tools/uwb_passive_ds_known_geometry_analyze.py
```

The complete machine-readable result is in `analysis_summary.json`; detailed methodology is in `README.md`.